

PROJECT OVERVIEW

AI Bathroom Designer

Take a photo of a bathroom. Get back a 3D design, a dimensioned floor plan and a price you could actually order from.

3 steps

Photo goes in, a priced design comes out.
Nothing in between needs a human.

200+

Vanities, toilets, showers, tubs, faucets and
finishes the engine can choose from.

2D + 3D

A CAD floor plan and a rendered room, both
showing the same fixtures.

The Problem, and What We Did About It

Most of a bathroom plan is spent checking things a computer can check instantly.

Today: the homeowner sketches something, the contractor says the door won't clear the toilet, the dealer quotes a number nobody budgeted for, and the whole thing starts again. Two or three weeks can disappear into that loop before anyone has agreed on a single fixture.

1

Describe it, or just photograph it

Type in the room size, your budget and the look you're after. If you'd rather not measure, upload a photo of the bathroom and the vision model estimates the dimensions, reads the existing style and finds where the plumbing already sits.

2

The engine does the checking

Every fixture in the catalog is tested against your room before it is ever scored: footprint plus clearance, door-swing radius, and whether the plumbing rough-in is reachable. Whatever survives gets ranked against what you said you wanted.

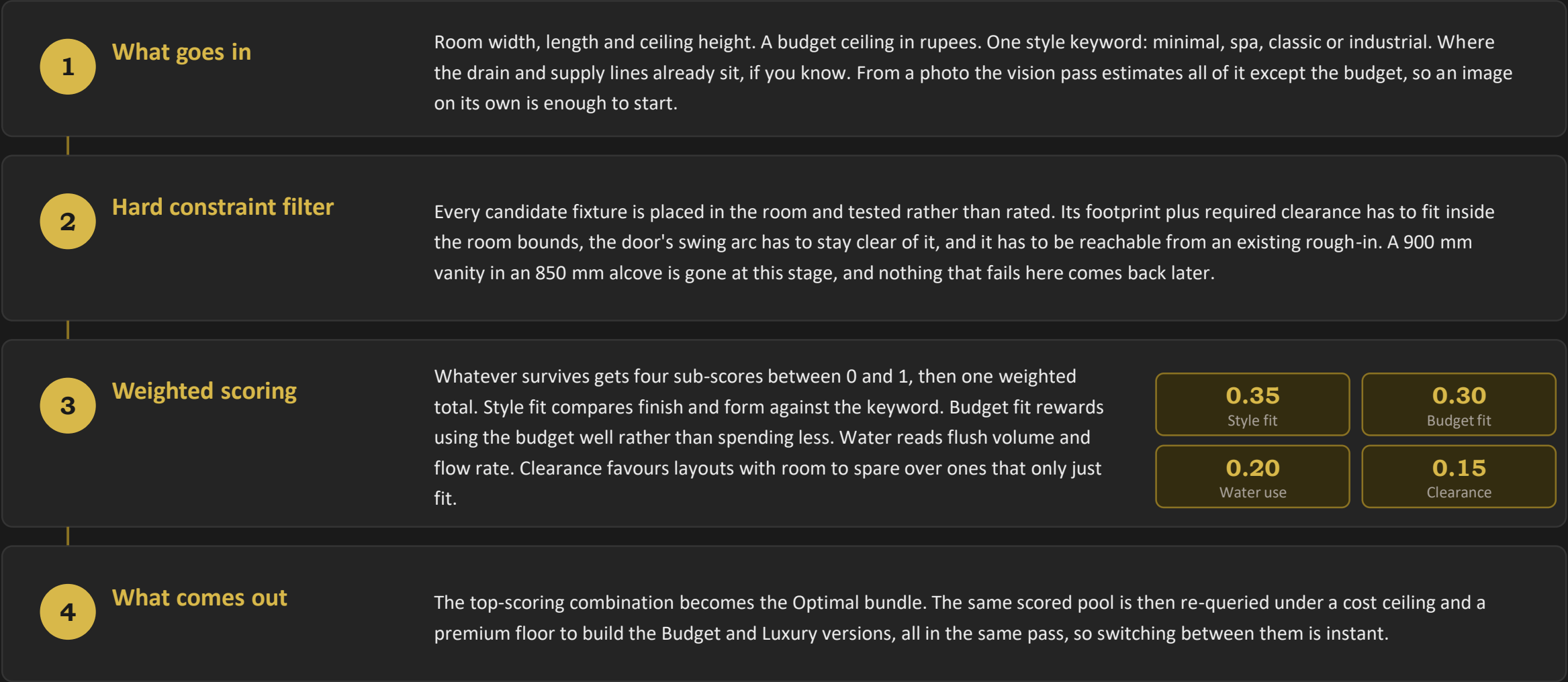
3

A design you can hand to a contractor

You get a rendered 3D room, a 2D plan with real dimensions, and three priced versions of the same layout. The bill of materials is GST-ready, so the quotation is the same document the fitter works from.

Inside the Recommendation Engine

What actually happens in the seconds between the photo and the design.



Three Versions of the Same Room

The engine doesn't ask you to choose a price bracket up front. It shows you all three.

Optimal

Balanced

The highest composite score across all four factors. This is the design the engine would pick if you gave it no further instruction.

- Sits close to the budget you set without drifting over
- Mid-range ceramics with the better tapware
- Widest clearance margins of the three

Budget Saver

About 30% less

The same layout re-costed under a hard ceiling. Savings come from fixture choice, not from cutting fixtures out of the room.

- Wall-hung and compact units to free up floor
- Single-piece ceramics instead of two-piece
- Drops the finishes first, the essentials last

Luxury Upgrade

About 40% more

Same footprint, better hardware. Useful when a client wants to see what the room could be before settling on what it will be.

- Smart toilets, rain towers, thermostatic mixers
- Stone or solid-surface counters
- Layered lighting and heated towel rails

What Makes It Different

Six things that separate this from a catalog with a 3D viewer bolted on.

1 Photo to design

One image is enough. The vision pass estimates room size, reads the existing style and locates the fixtures already in place, so a user who has never measured anything still gets a usable starting layout.

2 Constraints are enforced

Clearances, door swing and plumbing reach are a filter rather than a warning. If the engine returns a layout, it fits. There is no option to override it into something that doesn't.

3 Three tiers in one pass

Budget, optimal and luxury are produced together from the same scored pool. Changing your mind about spend costs nothing and takes no time.

4 A copilot that edits

Ask for a smaller vanity or a walk-in shower in plain language and the layout updates in place. It re-runs the same constraint checks, so a chat request can't produce an invalid room.

5 Quotation-ready output

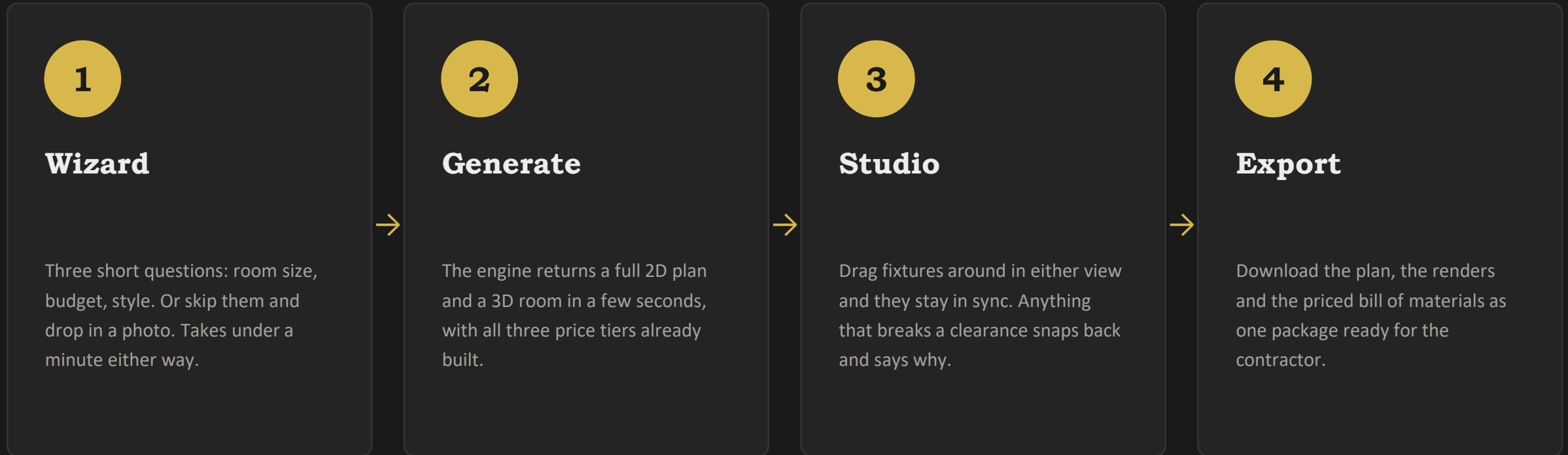
The bill of materials carries real SKUs, quantities and GST, which means the file the client approves is the file the fitter orders from.

6 Works without a paid key

The deterministic engine does the actual design work. Vision and chat are optional extras. With no API keys configured at all, the app still produces complete, priced designs.

What Using It Actually Looks Like

Four screens, start to finish.



The studio is where most of the time is spent. Everything before it exists to make sure the starting point is already close.

How It's Built

Chosen to stay fast, fully typed, and free to run.

FRONTEND

- React 18 and TypeScript, bundled with Vite
- Konva for the 2D CAD canvas; Three.js and React Three Fiber for the 3D room
- Tailwind and MUI for the interface, Zustand holding shared design state
- Voice and chat copilot wired into the same state, so edits show up in both views

BACKEND

- Express and TypeScript API serving recommendations and the bill of materials
- Deterministic constraint filter and weighted scorer, no model call required
- Anthropic or Gemini vision optional, used only for reading uploaded photos
- Runs end to end with zero paid API keys configured

IN SHORT

One tool that takes a room, or just a photo of one, and returns a design somebody can actually build and price.